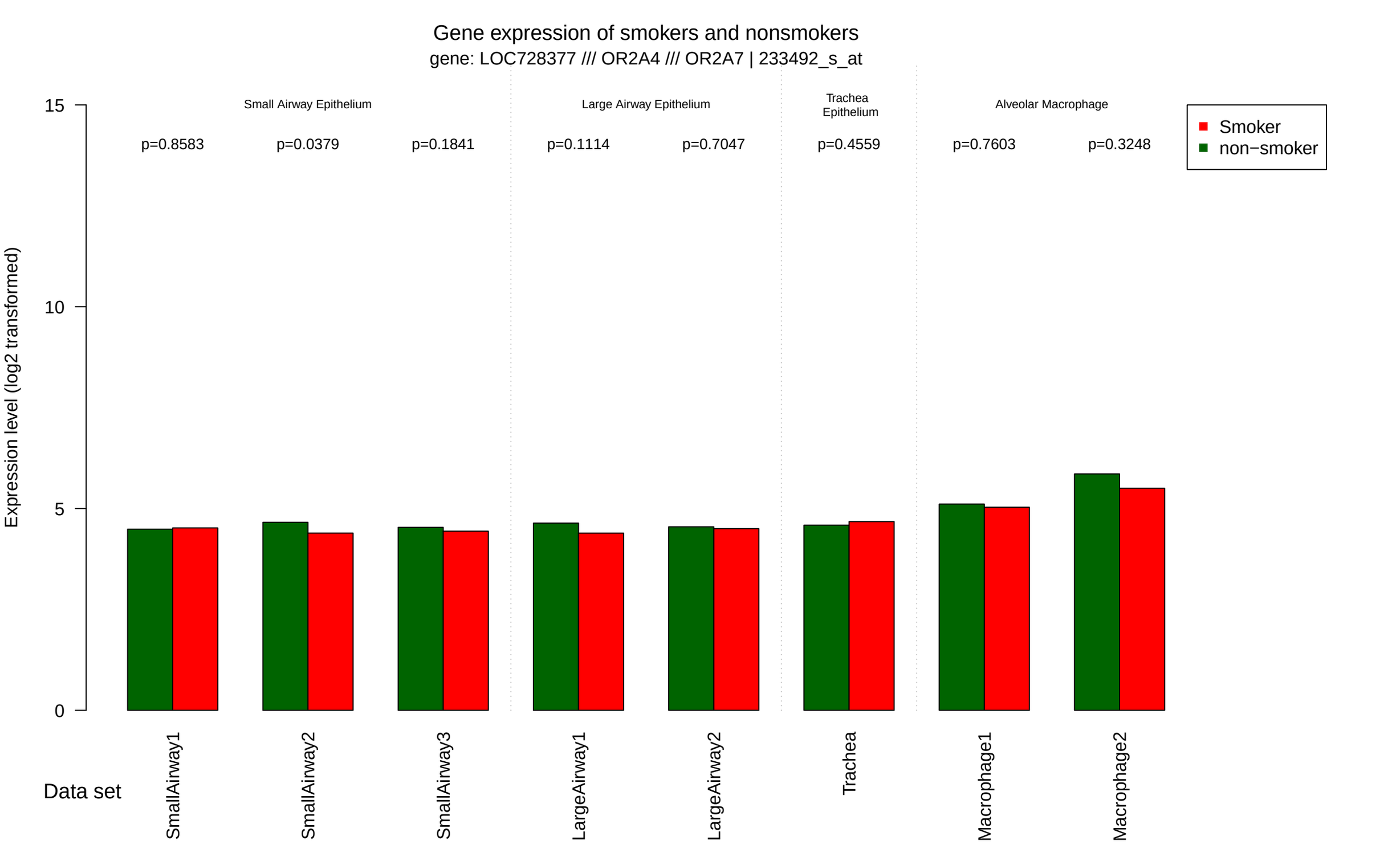
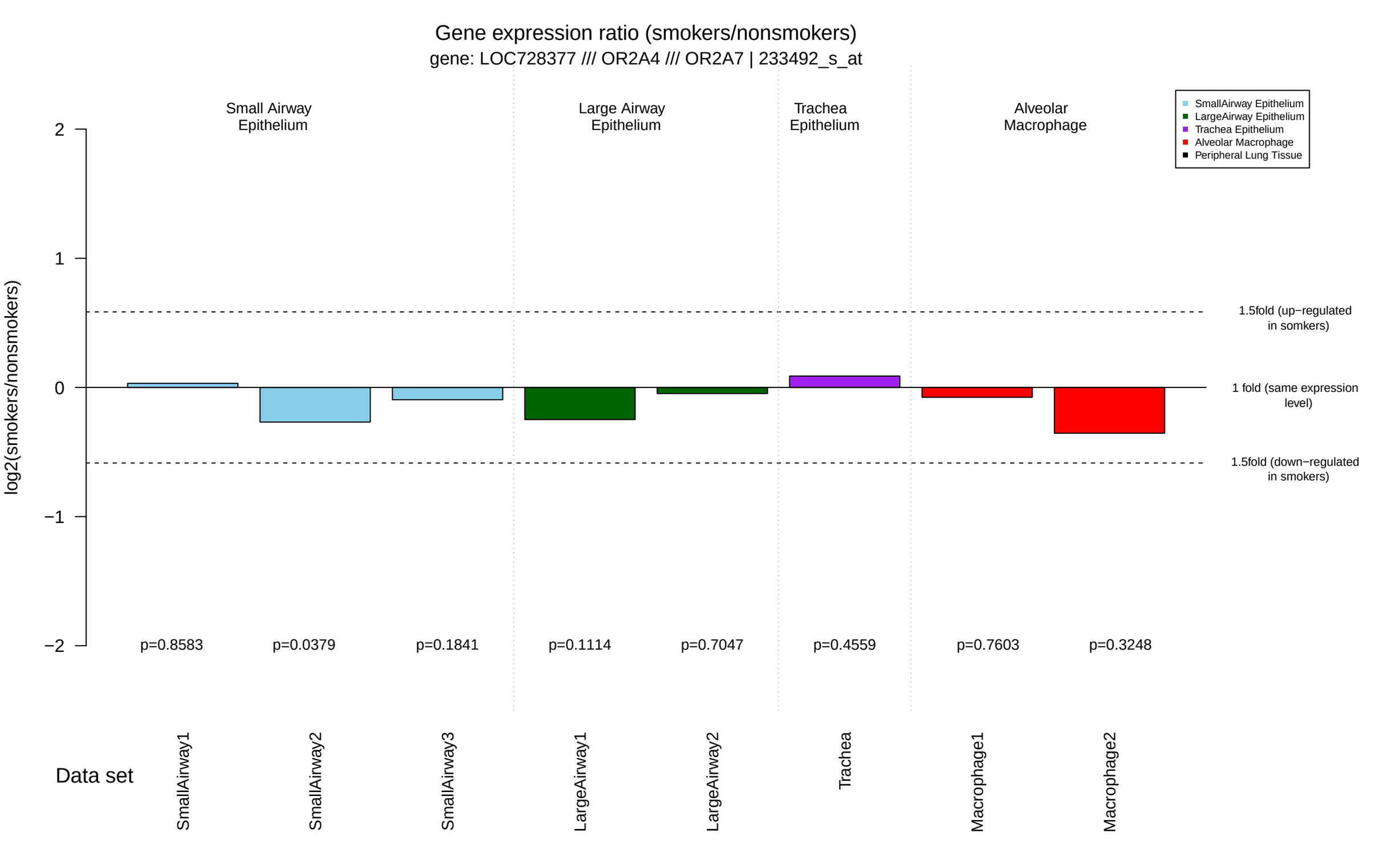
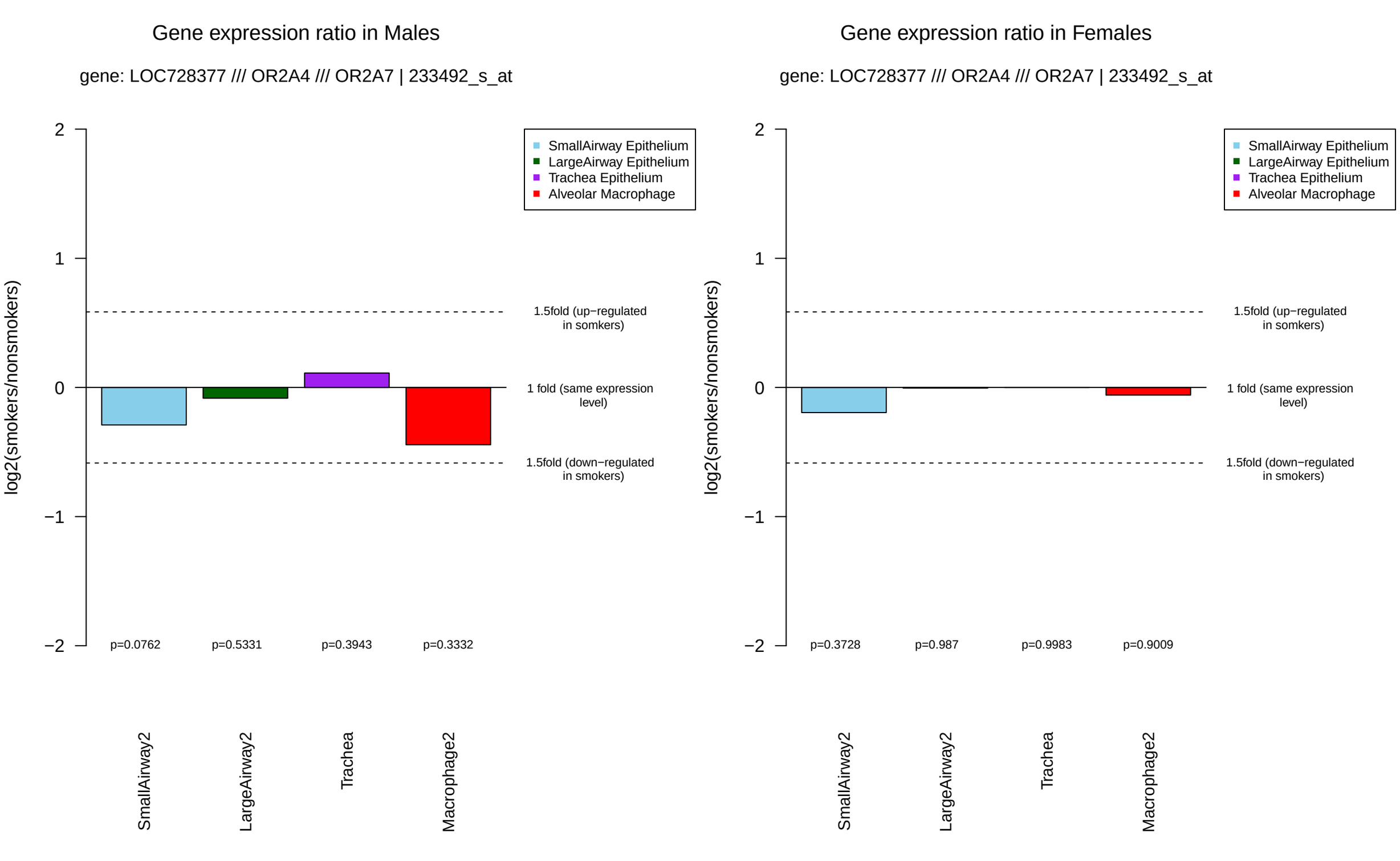
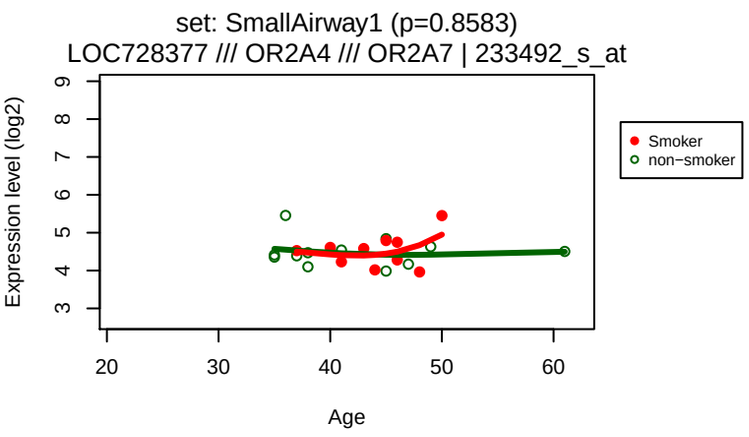
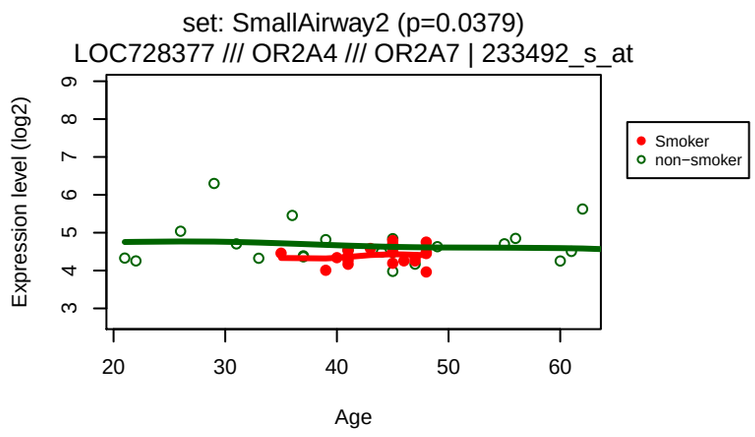
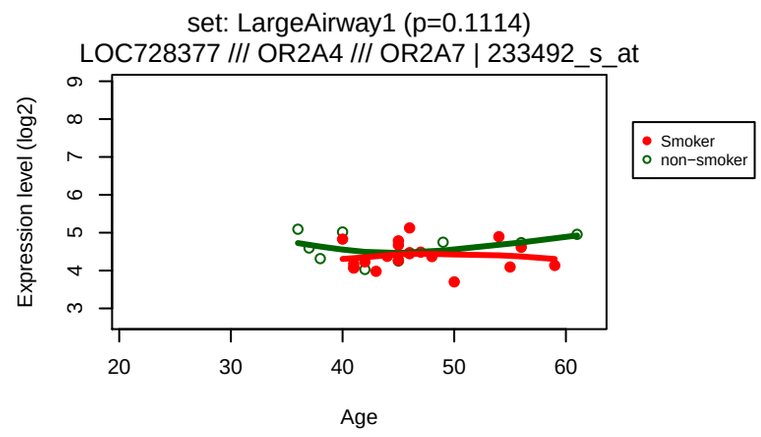

Gene expression of smokers and nonsmokers
gene: LOC728377 /// OR2A4 /// OR2A7 | 233492_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8583
p=0.0379
p=0.1841
p=0.1114
p=0.7047
p=0.4559
p=0.7603
p=0.3248
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

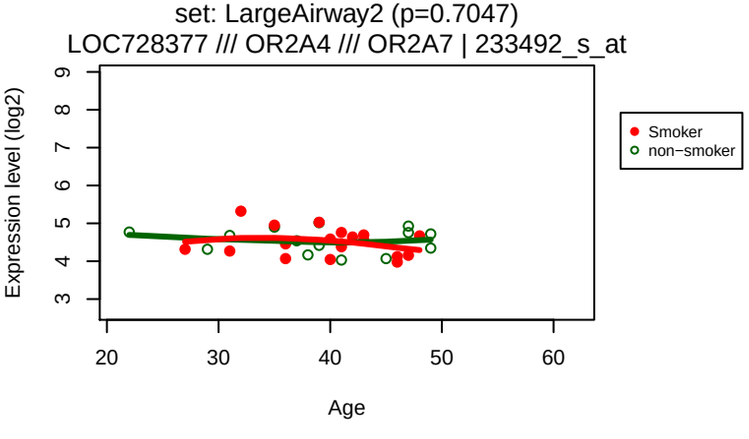

Gene expression ratio (smokers/nonsmokers)
gene: LOC728377 /// OR2A4 /// OR2A7 | 233492_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8583
p=0.0379
p=0.1841
p=0.1114
p=0.7047
p=0.4559
p=0.7603
p=0.3248
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

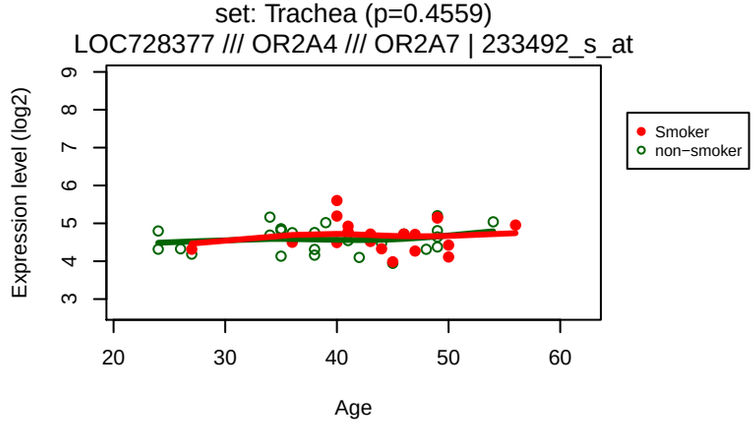

Gene expression ratio in Males
gene: LOC728377 /// OR2A4 /// OR2A7 | 233492_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0762
p=0.5331
p=0.3943
p=0.3332
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LOC728377 /// OR2A4 /// OR2A7 | 233492_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3728
p=0.987
p=0.9983
p=0.9009
SmallAirway2
LargeAirway2
Trachea
Macrophage2

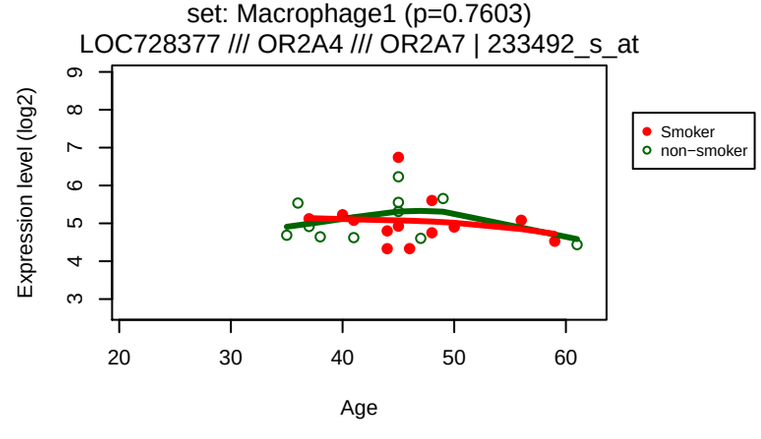

set: SmallAirway1 (p=0.8583)
LOC728377 /// OR2A4 /// OR2A7 | 233492_s_at
Expression level (log2)
Age
Smoker
non−smoker

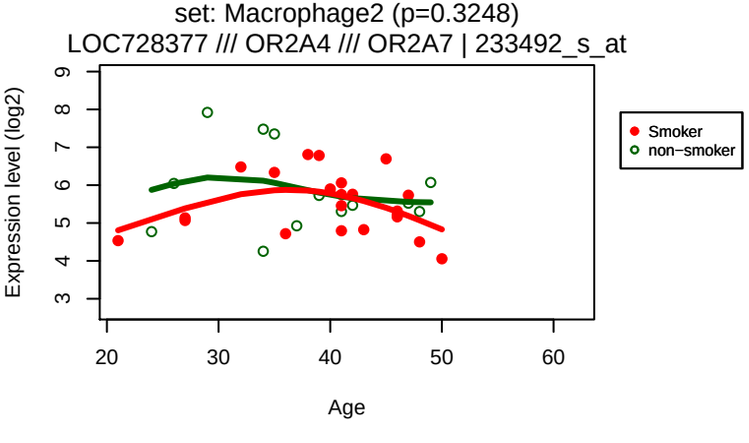

set: SmallAirway2 (p=0.0379)
LOC728377 /// OR2A4 /// OR2A7 | 233492_s_at
Expression level (log2)
Age
Smoker
non−smoker

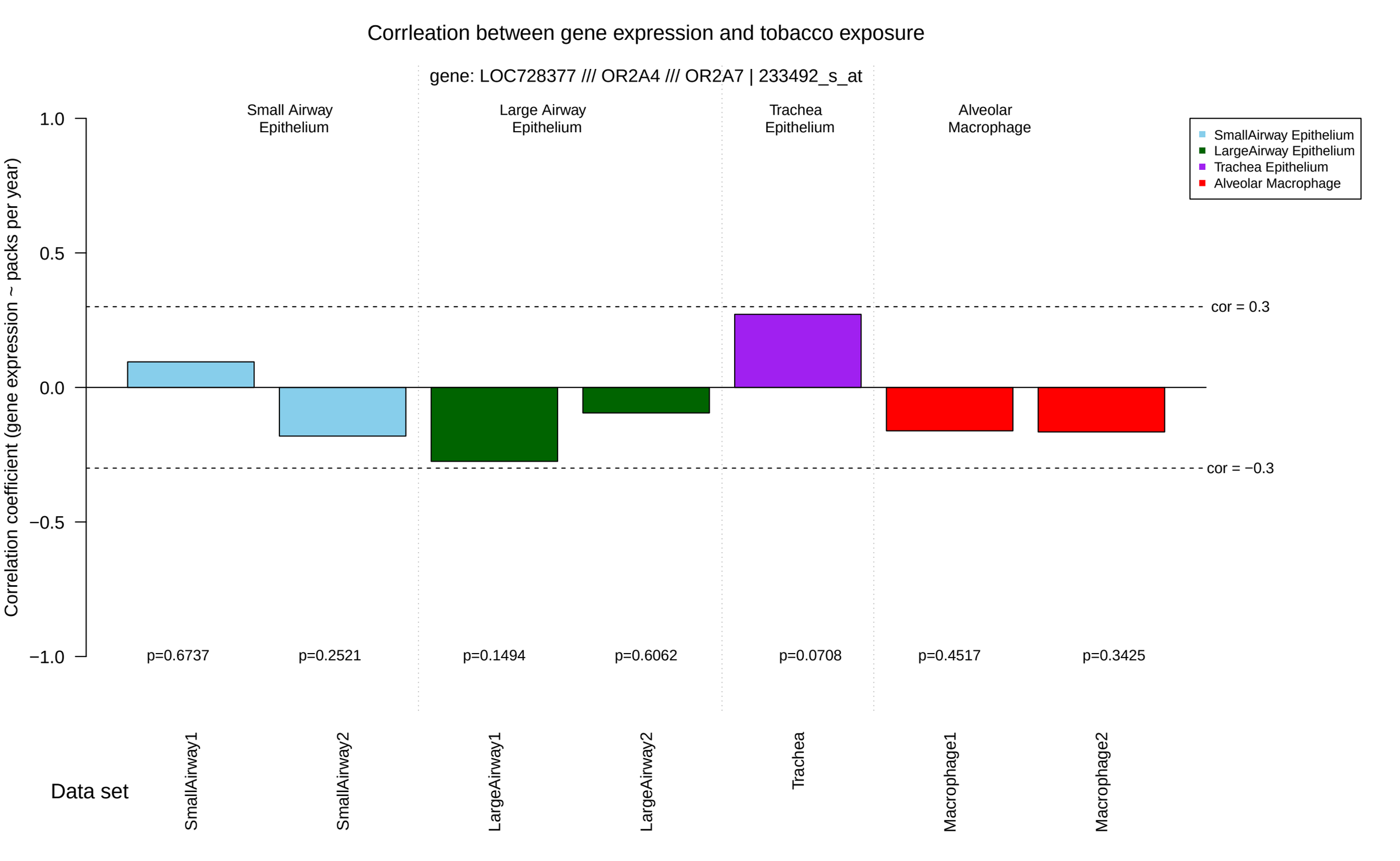

set: LargeAirway1 (p=0.1114)
LOC728377 /// OR2A4 /// OR2A7 | 233492_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7047)
LOC728377 /// OR2A4 /// OR2A7 | 233492_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4559)
LOC728377 /// OR2A4 /// OR2A7 | 233492_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7603)
LOC728377 /// OR2A4 /// OR2A7 | 233492_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3248)
LOC728377 /// OR2A4 /// OR2A7 | 233492_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LOC728377 /// OR2A4 /// OR2A7 | 233492_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6737
p=0.2521
p=0.1494
p=0.6062
p=0.0708
p=0.4517
p=0.3425
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2